Protocol / 2026-07-28

A compact message and trust reference.

Envelope

Request: jsonrpc="2.0", id, method, params. Response: same id with result or error.

Notification: method and params, no id and no response.

Client _meta: io.modelcontextprotocol/protocolVersion, clientInfo, clientCapabilities.

Ten core request methods

server/discover; tools/list; tools/call

resources/list; resources/templates/list; resources/read

prompts/list; prompts/get; completion/complete; subscriptions/listen

Results and caching

complete: inspect application outcome, including isError on tool results.

input_required: collect input, echo protected requestState, retry with inputResponses.

Cacheable results carry ttlMs and cacheScope. Keep private caches partitioned by identity.

Notifications

Progress and cancellation are request-correlated. Subscription acknowledgment starts an event stream.

Resource update and tool/resource/prompt list-change events invalidate cached state.

Protocol logging remains deprecated optional vocabulary; prefer stderr or OpenTelemetry.

Extensions

Tasks: resultType=task; tasks/get, tasks/update, tasks/cancel.

Skills: skills/list, skills/get; resources/read verifies bytes; directory reads are optional.

Apps have a separate UI bridge. OAuth exchanges and Server Cards are HTTP metadata, not new JSON-RPC methods.

Failures and trust

-32601 unknown method; -32602 invalid params; -32020 HTTP metadata mismatch; -32021 missing capability; -32022 unsupported revision.

Descriptions and schemas do not grant permission. Verify identity, scopes, arguments and exact approval independently.

Full captured examples: docs/PROTOCOL_GUIDE.md.